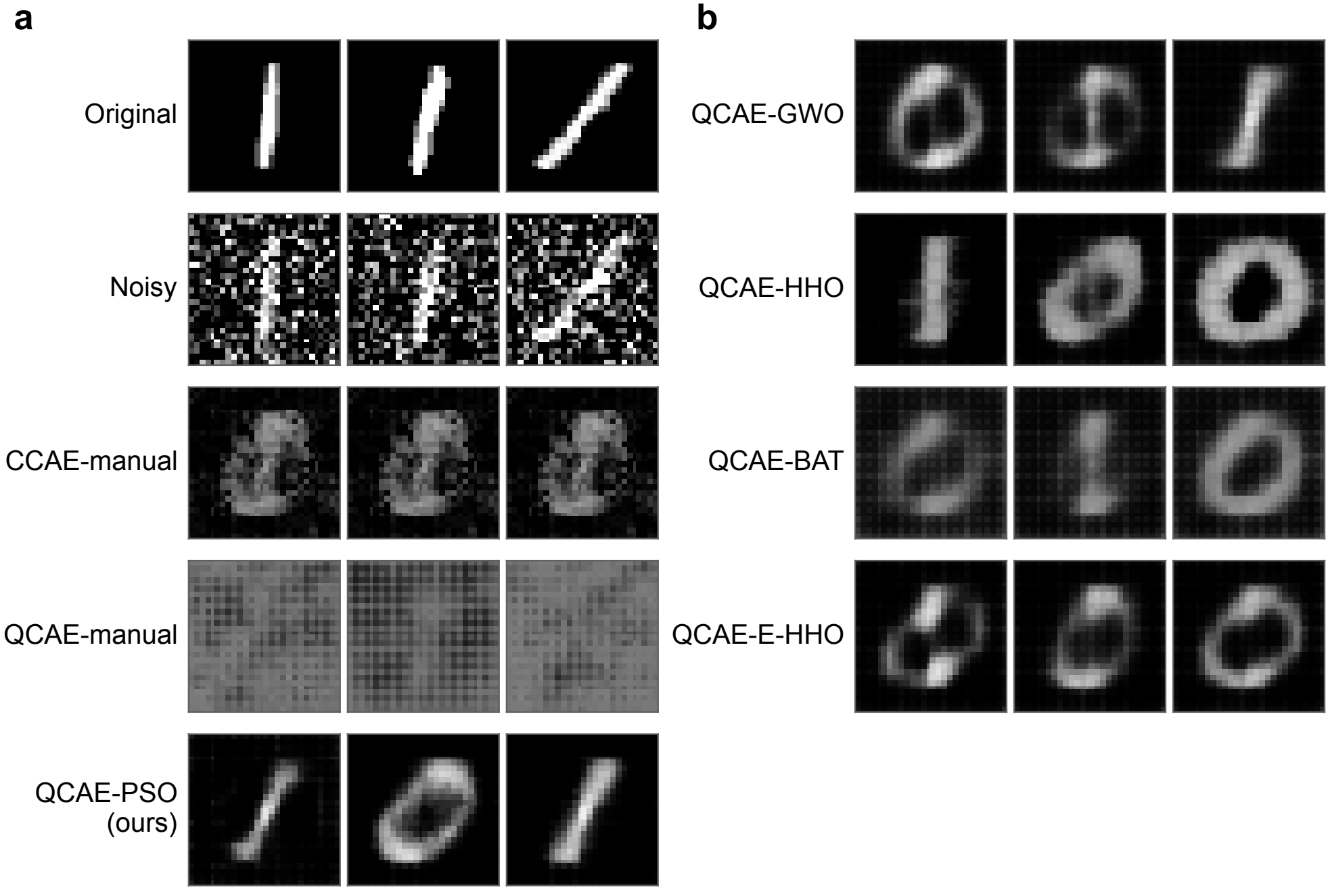

a
Original
Noisy
CCAE-manual
QCAE-manual
QCAE-PSO
(ours)
b
QCAE-GWO
QCAE-HHO
QCAE-BAT
QCAE-E-HHO